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Case Report

Tumefactive dilated perivascular space with spontaneous regression ☆

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ABSTRACT

Tumefactive (giant) perivascular spaces of the brain are a rare benign lesion that may occasionally result in localized neurological symptoms and typically do not change in appearance on subsequent imaging studies. Here we report a case of a 65-year-old woman with newly diagnosed lung cancer that underwent routine staging MRI of the brain. She was found to have a cystic temporal lobe lesion with imaging features consistent with a tumefactive perivascular space. Accurate diagnosis of this lesion was imperative in this case to avoid incorrect staging. On subsequent MRI, the lesion had markedly regressed without interval intervention further confirming the diagnosis. This case report presents the unique imaging features and presentation of this lesion to avoid unnecessary biopsy or resection.

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Introduction

Perivascular spaces are frequently seen on brain imaging studies and are fluid filled spaces that are continuous with subpial spaces surrounding the arterioles, venules and veins. These perivascular spaces occasionally become enlarged and when they are 15 mm or greater in size are referred to as tumefactive perivascular spaces (TPVS) [1]. TPVS is considered a rare lesion, can result in neurological symptoms and can mimic cystic tumors of the brain particularly when demonstrating peripheral hyperintensity on fluid sensitive MR sequences. Here we describe a rare case of an asymptomatic tumefactive perivascular space that spontaneously regressed on subsequent MR imaging.

Case summary

A 65-year-old very active woman with no smoking history developed progressive shortness of breath while running. Past medical history was only notable for hypothyroidism and she had no known family history of lung cancer. Laboratory studies were normal. She was initially diagnosed with pneumonia but unresponsive to antibiotics and subsequent CT of the chest showed a right lower lobe mass with mild increase SUV on PET CT (SUV 3.3) and no active lymph nodes. She underwent CT guided biopsy which showed a well-differentiated mucinous adenocarcinoma. Preoperative staging magnetic resonance imaging (MRI) of the brain demonstrated a left anterior temporal dilated subcortical perivascular space with

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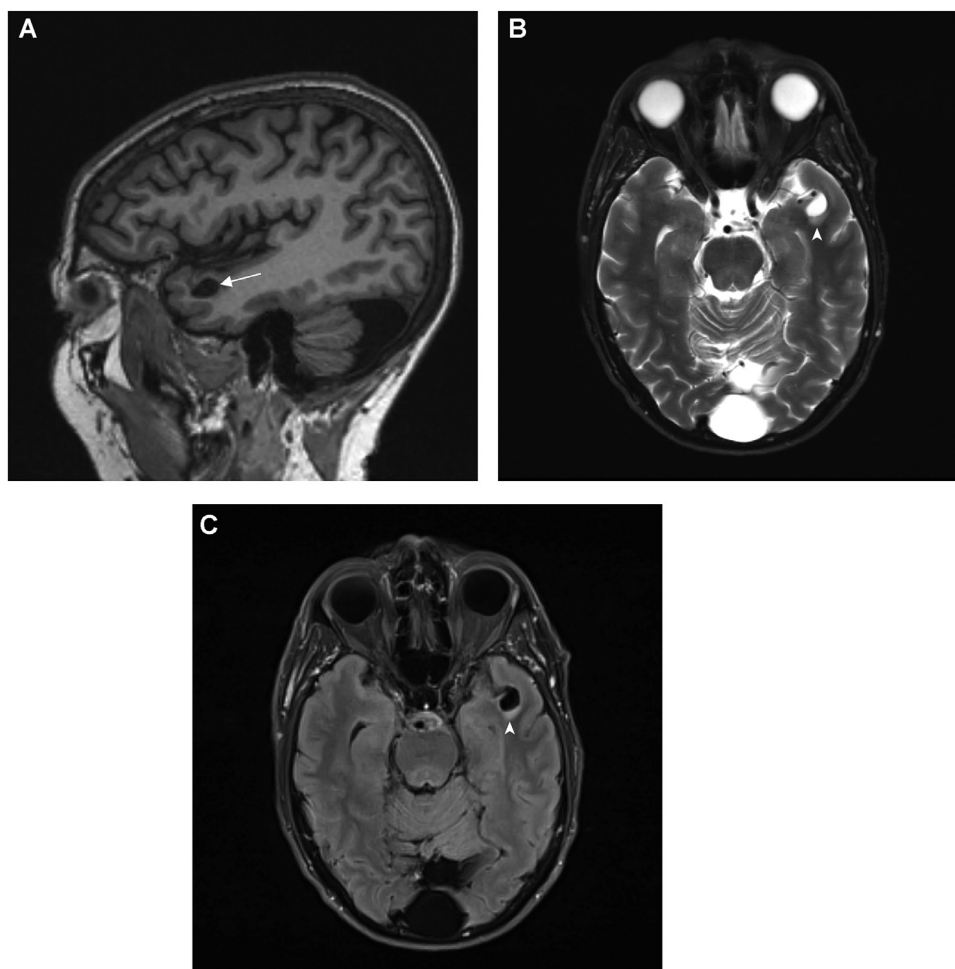


Fig. 1 – Sagittal T1 (A) MR image demonstrates a well-defined, unilocular, cystic lesion of the left temporal subcortical white matter (straight white arrow) with corresponding simple internal fluid intensity on axial T2 (B) and FLAIR (C) images with peripheral T2 and FLAIR hyperintensity surrounding the lesion (white arrowheads) thought to represent edema and/or gliosis. There was no enhancement of the lesion or adjacent white matter.

surrounding T2- and fluid-attenuated inversion recovery (FLAIR) hyperintense signal (figure 1) and no abnormal enhancement. This appearance was consistent with a tumefactive perivascular space. There was no MR evidence of intracranial metastatic disease. An arachnoid cyst of the superior-posterior tentorium cerebelli was noted. She subsequently underwent superior segment sparing thoroscopic right lower lobectomy and mediastinal lymphadenectomy which showed a well-differentiated mixed mucinous and nonmucinous adenocarcinoma with peribronchial invasion and positive regional lymph nodes (Stage IIIA: T2N2M0). She was started on routine chemotherapy with Cisplatin and Pemetrexed and regional radiation therapy. Patient remained asymptomatic and underwent subsequent 3-month surveillance MRI of the brain per treatment protocol that demonstrated marked regression of the perivascular space dilation and near resolution of the abnormal surrounding signal abnormality (Fig. 2).

Discussion

Perivascular spaces, also known as Virchow-Robin spaces, are normal anatomic structures of the subpial space composed of pia-mater lining that surround penetrating arterial and venous vessels supplying brain parenchyma [1–6]. Composed of interstitial fluid draining into the sub-arachnoid space through holes in the pia mater, perivascular spaces serve as a medium for exchange between the vasculature and adjacent brain parenchyma [1–6]. Typical perivascular spaces are tiny, smooth walled, and round to ovoid structures measuring less than 2 mm in diameter and isointense to CSF fluid on magnetic resonance imaging [1,2,4,5]. They involve the lenticulostriate arteries adjacent to the anterior commissure, though can also occur in penetrating arteries of the white matter and cerebral cortex [1,3–6].

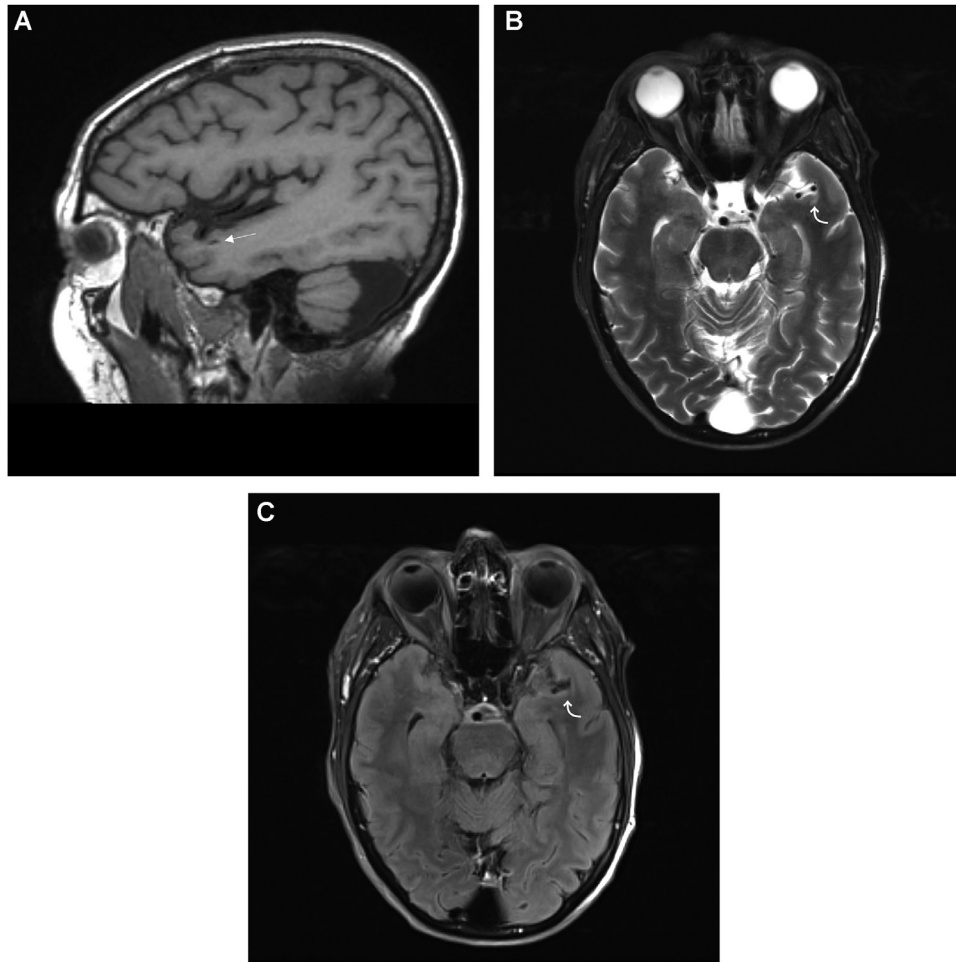


Fig. 2 – Sagittal T1 (A) MR image demonstrates near resolution of the cystic space (straight white arrow) with orientation consistent with a perivascular space and minimal residual T2 (B) and FLAIR (2C) peripheral hyperintensity (curved white arrows).

A perivascular space can become dilated and are considered TPVS, or giant perivascular spaces, if the diameter is 15 mm or greater and often exerts mass effect on local structures [1,5,6]. The penetrating vessels supplying the mesencephalothalamic region and cerebral white matter are the most common location for TPVS lesions [1,3–5]. TPVS are believed to result from trauma, ischemic injury, or neoplastic disease, which leads to the abnormal collection of interstitial fluid [1,4–6]. The exact underlying etiology of TPVS however remains unknown [1,2,4].

TPVS are rare and can be seen in patients of all ages and demographics, though it is more commonly associated with advancing age [1,2,4]. Spontaneous regression on follow up imaging is a very rarely reported event and to our knowledge only 7 patients have previously been reported, of which 6 had TPVS in the anterior temporal pole [4–7]. In one patient, regression followed resection of a nonfunctioning pituitary macroadenoma and another patient following resection of a convexity meningioma infiltrating the superior sagittal sinus [5,6]. One patient with 2 TPVS in both the anterior temporal pole and anterior perforate substance presented with hemorrhagic pituitary macroadenoma treated with medical

therapy with subsequent resolution of the pituitary lesion and anterior temporal lobe lesion while the anterior perforate substance TPVS remained unchanged [5]. These cases support the concept of obstruction of normal interstitial flow from the perivascular spaces being a plausible mechanism in development of TPVS. In our case however, the lesion regressed without intervention of the likely incidental arachnoid cyst of the tentorium cerebelli. There was also no apparent correlation with the standard chemotherapy regimen.

At time of presentation, patients are most commonly asymptomatic with TPVS being discovered incidentally on imaging, though those with focal mass effect may present with nonspecific symptoms, such as headache or dizziness [1–5]. This patient had no neurological symptoms. No specific treatment is recommended for management of TPVS, though intervention, such as CSF diversion, is pursued in cases of local mass effect resulting in obstructive hydrocephalus [3,7].

TPVS can be mistaken for alternative disease processes, most commonly cystic neoplasms [4,7]. Recognizing the characteristic appearance and location of TPVS can result in the correct diagnosis, which can avoid unnecessary surgical intervention if correctly identified [1,4]. This lesion should be

considered in patients with nonenhancing, cystic lesions exerting local mass effect that are isointense to CSF on all MRI sequences [1,3,4]. It can be unilocular or multilocular, and is found along the path of penetrating vessels [1,3]. Occasionally, thin peripheral hyperintense signal alterations in adjacent white matter can be seen on T2 and FLAIR sequences, attributed to either reactive vasogenic edema or gliosis secondary to mass effect causing advanced chronic ischemic changes or chronic mechanical stress from elevated arteriole pressure [1,4,5]. The limited nature of peripheral hyperintensity on MRI may suggest the diagnosis of TPVS as opposed to neoplasm. On computed tomography, the lesion shows hypodense CSF fluid attenuation and no enhancement following contrast administration [1].

Conclusion

Perivascular spaces are normal anatomic structures containing interstitial fluid around penetrating vessels supplying brain parenchyma. TPVS occurs when perivascular spaces become significantly dilated and may result in local mass. TPVS is infrequently reported in the literature, and even fewer cases have been reported with spontaneous regression but most often occur in the anterior temporal pole and may be associated with intracranial mass effect potentially disrupting normal interstitial flow of fluid into the CSF. The lesion has a characteristic appearance on imaging with a nonenhancing, smooth, and unilocular or multilocular cystic lesion that is iso-intense to CSF fluid on all MRI sequences with frequent limited peripheral FLAIR/T2 hyperintense signal. Differentiation of TPVS from cystic neoplasm is important to potentially avoid confusion with cystic neoplasm and potential unnecessary surgery.

Patient consent

Written informed consent for the publication of this manuscript has been obtained from the patient.

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